

MCR3M - M1C3

Agenda:

1. Transformations of Parabolas
2. Graphing Transformations
3. Portfolio Story #2
4. Consolidation: GoFormative Exit Card

Learning Goal:

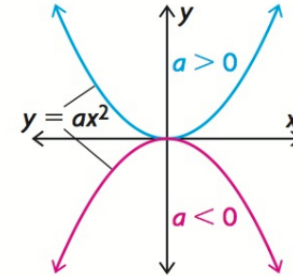
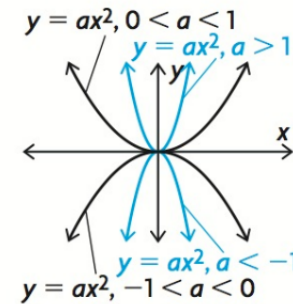
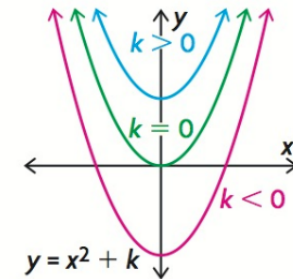
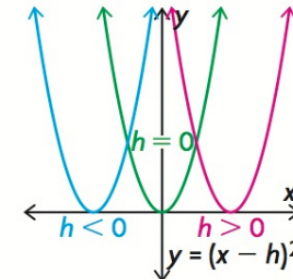
I will graph quadratic functions by applying transformations

Transformations

Use Desmos to explore how the parameters a , h and k affected the graph of a parabola from it's equation in **vertex form**:



- For a quadratic function in the form $g(x) = a(x - h)^2 + k$,
- horizontal translations:** The graph moves to the right when $h > 0$ and to the left when $h < 0$.
- vertical translations:** The graph moves up when $k > 0$ and down when $k < 0$.
- vertical stretches:** The graph is stretched vertically when $a > 1$.
- vertical compressions:** The graph is compressed vertically when $0 < a < 1$.
- vertical reflections:** The graph is reflected in the x-axis when $a < 0$.
- the axis of symmetry is the line $x = h$.
- the vertex is the point (h, k) .



Applying Transformations to Sketch Graphs

Ex. 1: Consider the function $g(x) = \frac{1}{4}(x + 3)^2 - 1$

- a) Identify the values of a, h, and k.

- b) Identify the transformations associated with each of the parameters above

You Try It:

Answer Formative questions 2 & 3 in M1C2

Applying Transformations to Sketch Graphs

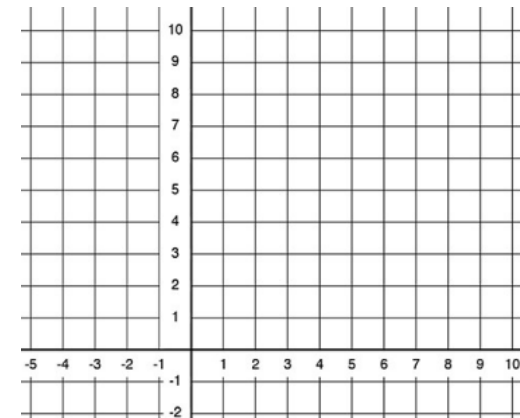
Transformations must always be applied in the following order:

1. Reflections/Stretches/Compressions (a)
2. Translations (h, k)

Ex. 2: Consider the function $f(x) = 2(x - 3)^2 + 1$

a) Identify the values of a, h, and k. What effect do these have on the parabola $y = x^2$?

b) Graph $f(x)$ by applying transformations to $y = x^2$. Show your work.

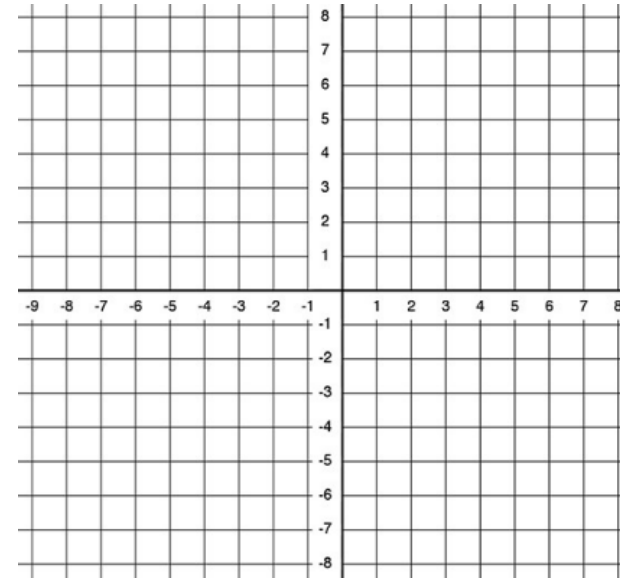


Applying Transformations to Sketch Graphs

Ex. 3: Consider the function $g(x) = -\frac{1}{2}(x - 1)^2$

a) Identify the values of a, h, and k. What effect do these have on the parabola $y = x^2$?

b) Graph $f(x)$ by applying transformations to $y = x^2$. Show your work.

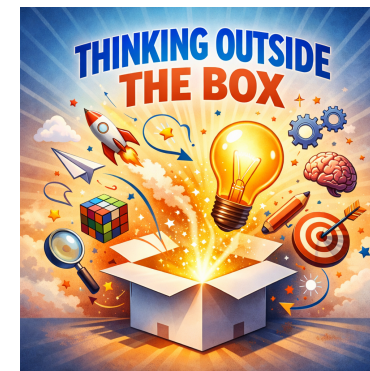


Story 2: Error Analysis, Verticiation, and Critical Thinking

Purpose: Show how you know when to trust an answer.

Required Components

- Choose **one (1) textbook question** (that requires 3+ steps) where you seriously checked your work. Include screenshots of the question and your original solution.
- Show at least **one method of verification** that was not required in the question, such as:
 - Substitution
 - Estimation
 - Graphing
 - An alternate algebraic method
 - AI
- If using AI to check your work
 - Include screenshots of your prompt and the AI output
 - Explain *how you evaluated its reasoning*, not just its final answer
 - If the AI made a mistake, clearly diagnose the error and fix it
- **Reflection**
 - How did I decide what to trust and what not to trust?
 - What signs told me an answer was correct or incorrect?



Cheat Code: Checked the textbook answer at least twice and still can't get there? That's not a problem — that's actually Story 2!

Find a more challenging homework problem you attempted - consider one toward the end of the recommended questions - and see if you can tackle it or verify it from another direction.

Can't think of a way to verify it on your own (via graphing, guess and check, etc.)? Attach an image of the question and your answer with the following AI prompt "*Check my answer here to verify if I'm correct or not. How could I have checked this on my own?*"

Consolidation

Complete question 4 of today's M1C2 Formative

Consider the function $h(x) = -2(x - 3)^2 + 1$

a) Identify the values of a, h, and k.

What effect do these have on the parabola $y = x^2$?

b) Graph $f(x)$ by applying transformations to $y = x^2$. Show your work (show table or curves to illustrate how points are affected).

